

CatNet: Triangulating Feature Structure in Adult Income Classification

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Abstract

CatNet (Complementary Analysis and Triangulated Network) is a reproducible six-stage analysis of the UCI Adult Census income classification task (Dua and Graff, 2019). Rather than treating the dataset only as a benchmark, we use it to examine how distributional shape, feature dependence, sparse predictive structure, and decision-boundary geometry relate to one another. The pipeline consists of **WhiskerFlow** for preprocessing, **KittenGauss** for class-conditional distribution analysis, **TailMI** for mutual-information diagnostics, **ClawReg** for L1-regularised logistic regression, **WhiskerInteract** for targeted interaction tests, and **PawSVM** for kernel boundary analysis. On the continuous variables, MeowRank scoring gives the largest marginal separation to education_num, age, and hours_per_week. TailMI shows a large dependency between marital_status and relationship, but WhiskerInteract does not find evidence that the tested numeric interactions improve prediction. ClawReg reaches a held-out ROC-AUC of 0.897 with a compact set of stable feature groups. Among the PawSVM variants, **PurrPoly** (degree-2 polynomial) reaches the highest held-out ROC-AUC at 0.905, while **FurRBF** (radial basis function) does not improve over **TabbySVM** (linear). We interpret this pattern as evidence for modest low-order nonlinearity rather than a need for highly flexible local decision boundaries. PurrRobust checks show that these conclusions are stable across random seeds, regularisation levels, and MI discretisation schemes.

1 Introduction

Since its introduction by Kohavi (1996), predicting whether an individual’s annual income exceeds \$50,000 from census records has become a standard binary classification task. The UCI Adult dataset (Dua and Graff, 2019), derived from the 1994 United States Census, combines continuous

demographic variables with categorical socioeconomic features and has moderate class imbalance. These properties make it useful not only for measuring predictive performance, but also for studying the structure that gives rise to that performance.

Most uses of this dataset emphasize accuracy, F1, fairness metrics, or comparisons among classifiers. We keep the same prediction task, but shift the emphasis toward interpretation. We ask four questions. First, what do the continuous feature distributions look like within each income class, and where is a Gaussian summary informative? Second, which features are statistically dependent, and do those dependencies translate into useful interaction terms? Third, which feature groups remain active under L1-enforced sparsity? Finally, is the income decision boundary close to linear, or does it contain meaningful low-order nonlinear structure?

To answer these questions, we use **CatNet**, a six-module workflow that keeps several kinds of evidence separate before comparing them. WhiskerFlow prepares the data. KittenGauss summarizes class-conditional continuous distributions. TailMI measures feature–label and feature–feature mutual information. ClawReg identifies sparse and stable predictors. WhiskerInteract tests selected numeric interactions. PawSVM compares linear, polynomial, and radial kernel boundaries. Reading these outputs together is useful because each view can fail in a different way. A feature may look strong marginally but add little once a correlated variable is present. A pair of features may have high mutual information because they encode the same social structure, not because their product is predictive.

The analysis leads to three main observations. Strong TailMI dependencies such as marital_status–relationship appear to reflect redundancy rather than useful numeric interactions. ClawReg suggests that much of the predictive signal is carried by a small and stable feature core. PawSVM results indicate that degree-2 PurrPoly

captures a small but consistent nonlinear gain, while FurRBF adds little over the linear TabbySVM baseline. PurrrRobust checks show that these observations are not sensitive to the particular random split, regularisation value, or MI discretisation scheme used in the main run.

2 System Design

CatNet connects six modules in a fixed order. Each module produces a small set of structured outputs, including figures, tables, and summary statistics, that are used to guide the next stage of analysis. All hyperparameter tuning is performed only on the training split. The following subsections describe the role of each module.

2.1 WhiskerFlow: Data Preprocessing

WhiskerFlow turns the raw UCI Adult records into the representation used by the later modules. The dataset contains 48,842 rows and combines continuous and categorical predictors. We remove two features before analysis: education, because it duplicates the ordinal information in education_num; and fnlwgt, following the project specification. The retained predictors include five continuous variables (age, education_num, capital_gain, capital_loss, hours_per_week) and seven categorical variables (workclass, marital_status, occupation, relationship, race, gender, native_country).

A key preprocessing step is the log-plus-one transform applied to capital_gain and capital_loss:

$$\tilde{x} = \log(1 + x), \quad x \geq 0. \quad (1)$$

This transform compresses the long right tails of these zero-inflated variables while preserving the point mass at zero. It produces capital_gain_log1p and capital_loss_log1p, which are used in ClawReg. We also collapse native_country into a binary grouped feature, native_country_grouped. This is appropriate for this analysis because approximately 89.7% of rows fall under the United States category, and the variable has very low feature-label MI (0.0006; see Section 5.3).

The main split is a stratified 80/20 train-test partition with random seed 42, yielding 39,073 training rows and 9,769 test rows. Stratification preserves the class ratio in both splits. For robustness checks, we repeat the same protocol with seeds 7, 42, and 99.

2.2 KittenGauss: Class-Conditional Distribution Analysis

KittenGauss fits a univariate Gaussian distribution to each continuous feature within each income class. The class-conditional model is $p(x | y = c) = \mathcal{N}(\mu_c, \sigma_c^2)$, where μ_c and σ_c are estimated from the training split by maximum likelihood.

We quantify single-feature separation in two ways. The **MeowRank** score is the held-out ROC-AUC obtained when the feature value alone is used as the classifier score. Cohen’s d (Cohen, 1988) measures the standardised mean difference between classes:

$$d = \frac{\bar{x}_{>50K} - \bar{x}_{\leq 50K}}{s_{\text{pooled}}}, \quad (2)$$

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}.$$

We use KittenGauss as a descriptive probe, not as a claim that the data are generated by Gaussians. The fitted curves help rank continuous features and make distributional failures visible, especially when a feature is sparse or zero-inflated.

2.3 TailMI: Feature Dependency Analysis

TailMI measures pairwise statistical dependence using empirical mutual information. For any pair of discrete or discretised variables X and Y , the estimate is:

$$\hat{I}(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \hat{p}(x, y) \log_2 \frac{\hat{p}(x, y)}{\hat{p}(x) \hat{p}(y)}, \quad (3)$$

where the joint and marginal distributions are estimated from training-split frequency counts. Continuous variables are discretised into equal-width bins before MI is computed. We use one coarse and one finer binning scheme to test whether the dependency ranking is sensitive to discretisation.

TailMI produces two outputs. The feature-label ranking identifies predictors with strong marginal association to income. The feature-feature matrix identifies pairs that share information and may therefore be redundant. Both outputs inform the WhiskerInteract candidate shortlist.

2.4 ClawReg: Sparse L1-Regularised Logistic Regression

ClawReg fits an L1-regularised logistic regression model, using the **ClawLoss** objective, across a grid

of regularisation strengths:

$$\begin{aligned} \mathcal{L}_{\text{Claw}}(\mathbf{w}; C) = & - \sum_{i=1}^n \left[y_i \log \sigma(\mathbf{w}^T \mathbf{x}_i) \right. \\ & \left. + (1 - y_i) \log(1 - \sigma(\mathbf{w}^T \mathbf{x}_i)) \right] \\ & + \frac{1}{C} \|\mathbf{w}\|_1, \end{aligned} \quad (4)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the sigmoid function and $C > 0$ controls inverse regularisation strength. The regularisation grid spans nine logarithmically spaced values from $C = 0.001$ to $C = 10.0$. We select C by 5-fold cross-validated ROC-AUC on the training split. To study stability, we record the fraction of grid points at which each feature group has at least one non-zero coefficient.

2.5 WhiskerInteract: Targeted Interaction Validation

WhiskerInteract tests a pre-specified set of numeric interaction candidates against the main-effects ClawReg model. We shortlist candidates using the TailMI feature–feature matrix and simple domain plausibility, and we restrict the search to numeric pairs to avoid the combinatorial growth of categorical interactions. For each candidate, we append the product of the two standardised predictors to the feature matrix and refit the ClawLoss model. We treat an interaction as potentially useful only if it improves cross-validated ROC-AUC by more than 0.001 over the main-effects baseline.

2.6 PawSVM: Kernel Boundary Shape Analysis

PawSVM compares three SVM variants as a way to probe the shape of the income decision boundary. All variants solve the soft-margin SVM primal:

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & y_i (\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \end{aligned} \quad (5)$$

with different kernel functions $k(\mathbf{x}_i, \mathbf{x}_j)$. **TabbySVM** uses a linear kernel. **PurrPoly** uses a degree- d polynomial kernel:

$$k_{\text{PurrPoly}}(\mathbf{x}_i, \mathbf{x}_j) = (\gamma \mathbf{x}_i^T \mathbf{x}_j + r)^d. \quad (6)$$

FurRBF uses a radial basis function kernel:

$$k_{\text{FurRBF}}(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2). \quad (7)$$

Each variant is tuned independently by grid search on the training split using 5-fold cross-validated ROC-AUC. We then compare the tuned kernels on held-out test ROC-AUC.

3 Datasets Used

The UCI Adult dataset was introduced as a machine learning benchmark by [Kohavi \(1996\)](#). It has since been widely used to evaluate classifier performance, fairness interventions, and interpretability methods. Several preprocessing choices have become common, including removing the `fnlwgt` sample weight and replacing the string-valued education field with the ordinal `education_num` variable. In this work, we follow these conventions but treat the dataset itself as the object of analysis.

Mutual information provides a non-parametric measure of statistical dependence ([Shannon, 1948](#); [Cover and Thomas, 2006](#)). Unlike linear correlation, it can capture arbitrary functional relationships. This makes it useful for detecting nonlinear feature–label associations and for identifying redundancy between feature pairs. Filter-based feature selection methods based on MI have been studied extensively ([Hastie et al., 2009](#)). Here, TailMI uses MI mainly as a diagnostic tool rather than as a direct feature-selection rule.

L1-regularised logistic regression, popularised through the LASSO by [Tibshirani \(1996\)](#), produces exact coefficient sparsity and is often used to identify compact sets of informative features. [Andrew and Gao \(2007\)](#) describe efficient algorithms for L1-regularised log-linear models. ClawReg studies the full regularisation path, which lets us distinguish feature groups that remain active under strong sparsity from those that appear only when regularisation is weak.

Support vector machines ([Cortes and Vapnik, 1995](#)), combined with kernel methods ([Boser et al., 1992](#)), provide a direct way to compare assumptions about decision-boundary shape. A linear kernel asks whether the encoded features are sufficient under an approximately linear separator. Polynomial and radial basis function kernels allow progressively more flexible boundaries. We use ROC-AUC as the primary metric because the dataset is imbalanced and because ROC-AUC evaluates ranking quality across probability thresholds ([Fawcett, 2006](#); [Hand, 2009](#)). All implementations use scikit-learn ([Pedregosa et al., 2011](#)).

4 Methodology

We use the WhiskerFlow 80/20 stratified train–test split with seed 42 as the primary evaluation setting. Hyperparameters are tuned only on the training split with 5-fold stratified cross-validation. The test split is reserved for final evaluation and robustness reruns, and is not used for model selection.

Before fitting ClawReg and PawSVM, categorical features are one-hot encoded and continuous features are standardised with training-split means and variances. The TabbySVM grid uses $C \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$. PurrPoly uses $d \in \{2, 3\}$, $C \in \{0.1, 1.0, 10.0\}$, and $r \in \{0.0, 1.0\}$. FurRBF uses $C \in \{1.0, 3.0, 10.0\}$ and $\gamma \in \{0.01, 0.05, 0.1, 0.5\}$. We report accuracy, macro-averaged precision and recall, positive-class F1, and ROC-AUC. ROC-AUC is used as the main comparison metric. All implementations use scikit-learn 1.8.0 (Pedregosa et al., 2011) with Python 3.12.

5 Results

5.1 Focused Exploratory Analysis

The training split contains 39,073 rows. Of these, 29,724 (76.1%) belong to the $\leq 50K$ class and 9,349 (23.9%) belong to the $> 50K$ class. This imbalance, shown in Figure 1, motivates the use of threshold-aware metrics such as F1 alongside threshold-independent ROC-AUC.

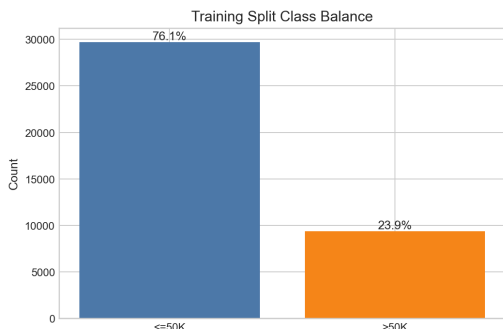


Figure 1: Class distribution in the training split. The $> 50K$ class constitutes 23.9% of training instances, which motivates reporting ROC-AUC alongside threshold-dependent metrics.

The continuous features show several patterns that matter for interpretation. `capital_gain` is highly sparse, with 91.8% zeros, and `capital_loss` is even sparser, with 95.2% zeros. `hours_per_week` also has a strong repeated value, with 46.5% of rows taking one value. These pat-

terns explain why the WhiskerFlow log1p transforms are useful, and they also shape how we interpret the KittenGauss fits. Figure 2 shows visible class separation in `education_num` and `age`, in contrast to the zero-dominated capital variables.

The categorical variables also show large differences in positive-income rates across groups. Figure 3 highlights three examples: `Married-civ-spouse` has a 44.8% positive rate compared with 4.4% for `Never-married`; `Exec-managerial` has 48.2% compared with 4.0% for `Other-service`; and `Male` has 30.4% compared with 10.9% for `Female`. These contrasts suggest, before formal modelling, that household structure and occupation-related variables will carry substantial information about the label.

5.2 KittenGauss Distribution Analysis

Table 1 reports the class-conditional Gaussian statistics and MeowRank scores for the five continuous features. `education_num` has the largest continuous separation under this single-feature view, with a MeowRank ROC-AUC of 0.716 and a Cohen’s d of 0.825. `age` and `hours_per_week` show more moderate separation. The capital variables have identical class medians of zero, which is consistent with their extreme zero inflation.

Figure 4 overlays empirical histograms with the fitted class-conditional Gaussian curves. The approximation is reasonable for `education_num`, `age`, and `hours_per_week`. It breaks down for `capital_gain` and `capital_loss`, where the distribution is mostly zero with a sparse right tail. Figure 5 ranks the five features by MeowRank score and again places `education_num` first among the continuous variables.

5.3 TailMI Feature Dependencies

Table 2 reports the largest feature–label and feature–feature MI values. The highest feature–label MI values are for `relationship` (0.116) and `marital_status` (0.111). These are above the leading numeric features, including `age` (0.064) and `education_num` (0.062). The largest pairwise dependencies are concentrated among household-structure and work-context variables: `marital_status`–`relationship` ($\hat{I} = 0.725$), `workclass`–`occupation` (0.329), and `relationship`–`gender` (0.270).

Figure 6 shows the full pairwise MI matrix. Figures 7 and 8 show the feature–label ranking and the leading feature–feature pairs as bar charts. The dis-

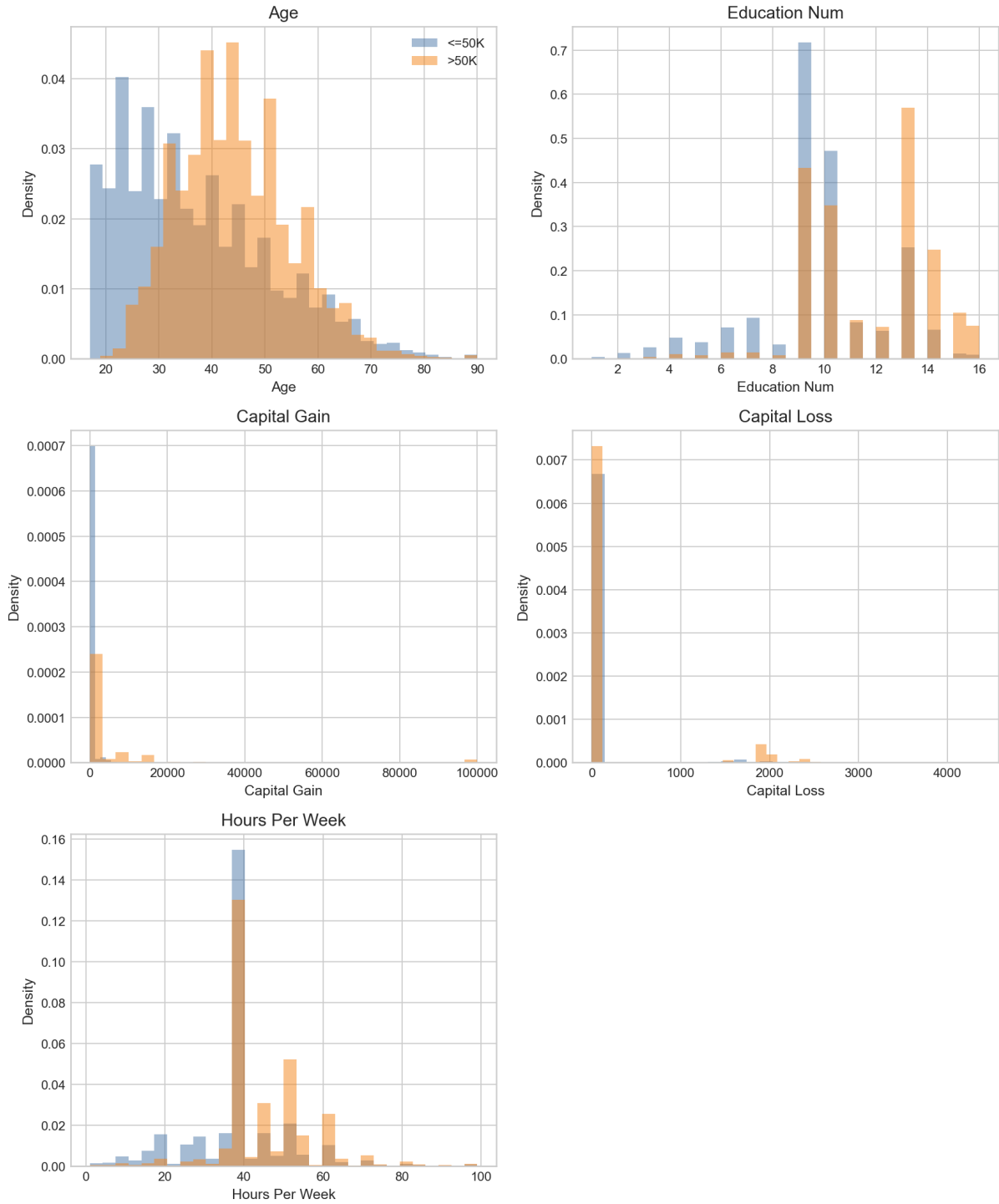


Figure 2: Continuous feature distributions by income class. education_num, age, and hours_per_week show visible class separation. capital_gain and capital_loss are dominated by a zero mass and long right tails, so a Gaussian summary captures their shape poorly.

cretisation sensitivity check gives complete overlap in the top-10 MI pairs (10/10) and in the candidate interaction shortlist (5/5) across the two binning schemes. This suggests that the main TailMI dependency pattern is not an artefact of the chosen bin width.

5.4 ClawReg Sparse Model

Cross-validation selects $C = 1.0$ for ClawReg. Table 3 reports the held-out performance of this model, including a test ROC-AUC of 0.897.

The cross-validated ROC-AUC rises from 0.887 at $C = 0.001$ to approximately 0.900 by $C = 0.1$.

Feature	Mean ($\leq 50K$)	Mean ($> 50K$)	Cohen's d	MeowRank AUC
education_num	9.60	11.60	0.825	0.716
age	36.92	44.35	0.556	0.682
hours_per_week	38.89	45.43	0.541	0.671
capital_gain	147.7	3950	0.532	0.588
capital_loss	55.3	200.0	0.358	0.535

Table 1: KittenGauss statistics for the five continuous features. MeowRank AUC is the single-feature held-out ROC-AUC. The capital variables have large mean differences but low MeowRank scores, which is consistent with their zero-inflated shape.

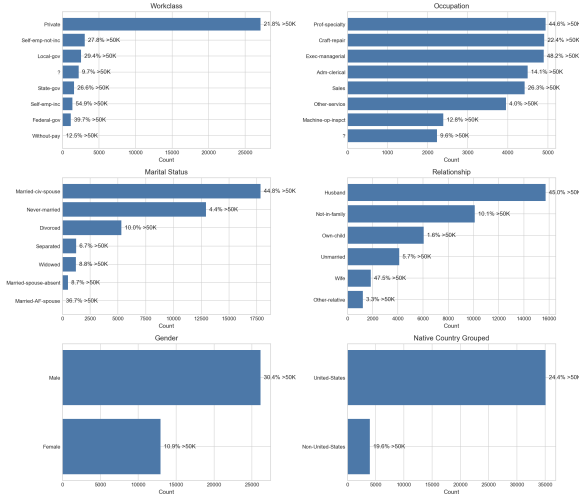


Figure 3: Selected categorical feature distributions with positive-income rates. Household-structure variables (marital_status, relationship) and occupation-related variables show large income-class differences before formal modelling.

Feature	Feature-label MI
relationship	0.116
marital_status	0.111
age	0.064
occupation	0.063
education_num	0.062
capital_gain	0.054

Feature pair	Pairwise MI
marital_status $\times$ relationship	0.725
workclass $\times$ occupation	0.329
relationship $\times$ gender	0.270
age $\times$ marital_status	0.233
education_num $\times$ occupation	0.213

Table 2: TailMI results. *Top*: top feature-label MI values. *Bottom*: top five feature-feature MI pairs. The high pairwise MI values suggest structural overlap among sociodemographic variables.

After that point, the curve is nearly flat. From $C = 0.1$ to $C = 10.0$, mean CV ROC-AUC varies by only 0.0003 (0.8998, 0.9001, 0.9001, and 0.9001 at $C = 0.1, 0.316, 1.0, 10.0$). This suggests that

Acc.	Prec.	Rec.	F1	AUC	Best C
0.848	0.723	0.589	0.649	0.897	1.0

Table 3: ClawReg held-out performance on the test split (seed 42). Acc. = accuracy, Prec. = precision, Rec. = recall, AUC = ROC-AUC.

the logistic model is not very sensitive to the exact regularisation level once moderate sparsity is allowed.

Table 4 reports feature groups that remain non-zero across the ClawLoss regularisation grid. Six groups are active at all nine grid points: marital_status, education_num, capital_gain_log1p, hours_per_week, age, and capital_loss_log1p. Among these, marital_status has the largest mean absolute coefficient, 3.359, followed by education_num at 0.741.

Feature group	Non-zero share	Mean coef
marital_status	1.000	3.359
education_num	1.000	0.741
capital_gain_log1p	1.000	0.475
hours_per_week	1.000	0.328
age	1.000	0.317
capital_loss_log1p	1.000	0.220
occupation	0.889	—
relationship	0.778	—
workclass	0.778	—
gender	0.778	—

Table 4: ClawReg feature group stability across the full regularisation grid. Non-zero share is the fraction of the 9 grid points at which the group has at least one non-zero coefficient. Groups with non-zero share < 1 do not appear at all grid points.

Figure 9 shows the ClawLoss coefficient paths across the regularisation grid. The stable groups remain active across the full path, while several peripheral groups enter only as regularisation weakens. We interpret this as evidence that the model relies on a compact feature core, with additional

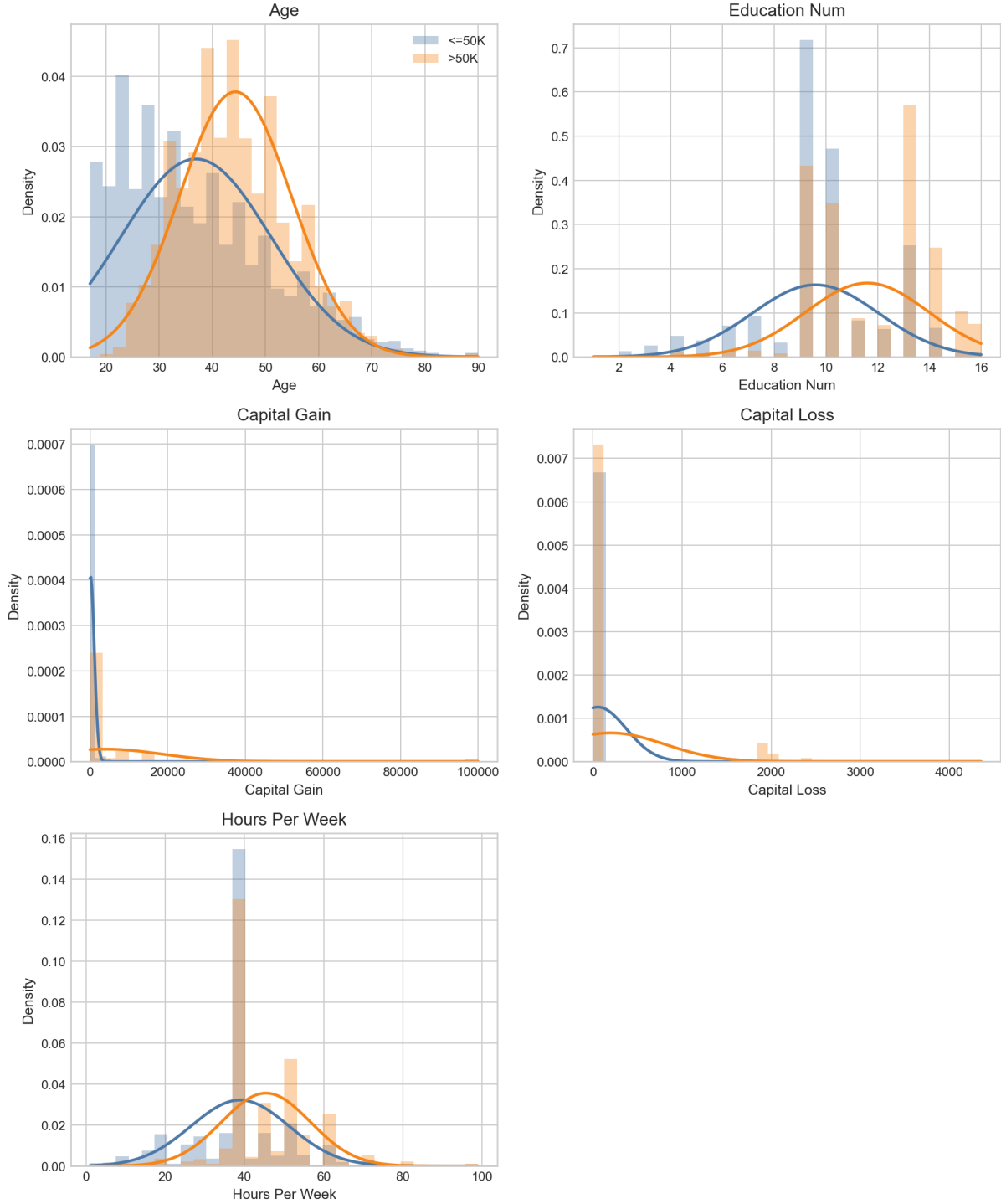


Figure 4: KittenGauss fitted overlays. Empirical histograms are shown with class-conditional Gaussian curves for each continuous feature. The fit is reasonable for education_num, age, and hours_per_week, but is a poor description of the zero-inflated capital variables.

groups contributing smaller or less stable increments.

5.5 WhiskerInteract Interaction Validation

We next use WhiskerInteract to test five numeric interaction candidates drawn from the TailMI short-list. Table 5 summarises the results. Every

tested interaction changes cross-validated ROC-AUC by less than 10^{-3} in magnitude. The largest positive CV change is $+0.000174$ for $\text{age} \times \text{hours_per_week}$. None of the tested interactions improves both cross-validation and test ROC-AUC in a consistent way.

Interaction	Pairwise MI	Δ CV AUC	Δ Test AUC
age $\times$ hours_per_week	0.066	+0.000174	+0.000210
age $\times$ education_num	0.052	−0.000026	−0.000012
education_num $\times$ hours_per_week	0.027	−0.000043	−0.000003
education_num $\times$ capital_gain	0.016	−0.000025	+0.000057
capital_gain $\times$ hours_per_week	0.008	+0.000042	−0.000066

Table 5: WhiskerInteract results. All ROC-AUC deltas fall below the 0.001 improvement threshold, so no interaction-augmented model was retained.

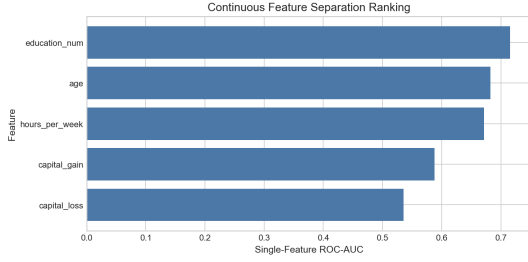


Figure 5: MeowRank ranking of continuous features by single-feature held-out ROC-AUC. education_num has the largest marginal separation under this metric.

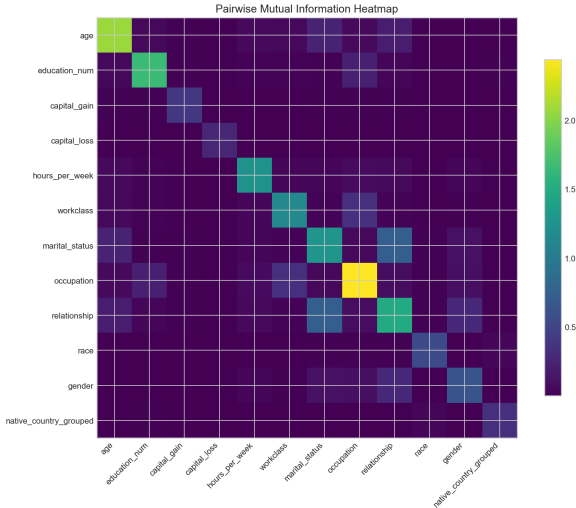


Figure 6: TailMI pairwise mutual information heatmap. The largest dependencies are concentrated among household-structure and work-context variables. Darker cells indicate higher MI values.

Because no interaction clears the 0.001 threshold, and because the test-set delta signs vary across candidates, we do not retain an interaction-augmented final model. Figure 10 shows the same pattern visually.

5.6 PawSVM Boundary Analysis

Table 6 compares the three PawSVM variants. TabbySVM reaches a test ROC-AUC of 0.897 at $C = 0.1$. PurrPoly with degree 2 reaches 0.905, about 0.008 higher. FurRBF reaches 0.896, slightly

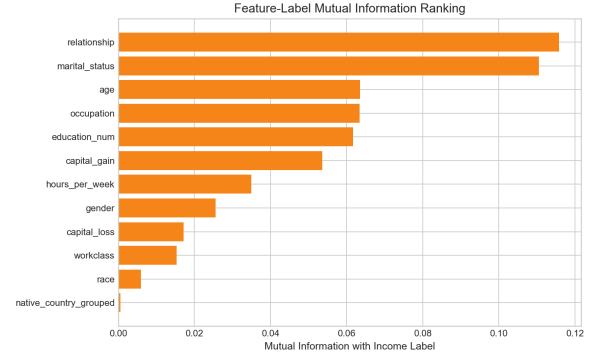


Figure 7: TailMI feature-label MI ranking. relationship and marital_status have the largest marginal MI with the income label.

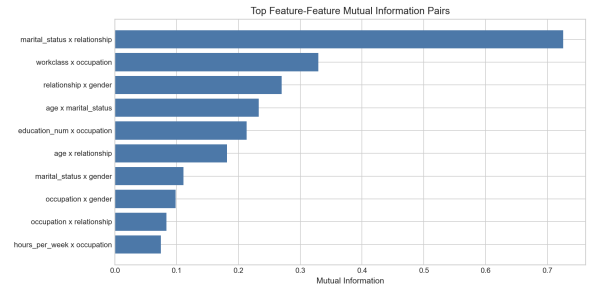


Figure 8: Top feature-feature MI pairs. The largest values mostly reflect overlap among sociodemographic variables.

below TabbySVM.

The degree-3 PurrPoly variant performs worse in cross-validation (CV ROC-AUC $\approx$ 0.883) than degree 2 (0.898). This suggests that the useful nonlinear signal is low-order. The FurRBF result is also consistent with the WhiskerInteract results: the dataset does not appear to require a highly flexible local boundary. One possible explanation for the PurrPoly gain is that the degree-2 kernel captures many small second-order effects across the full encoded feature space, rather than a small set of large numeric products. Figure 11 summarises the held-out ROC-AUC comparison.

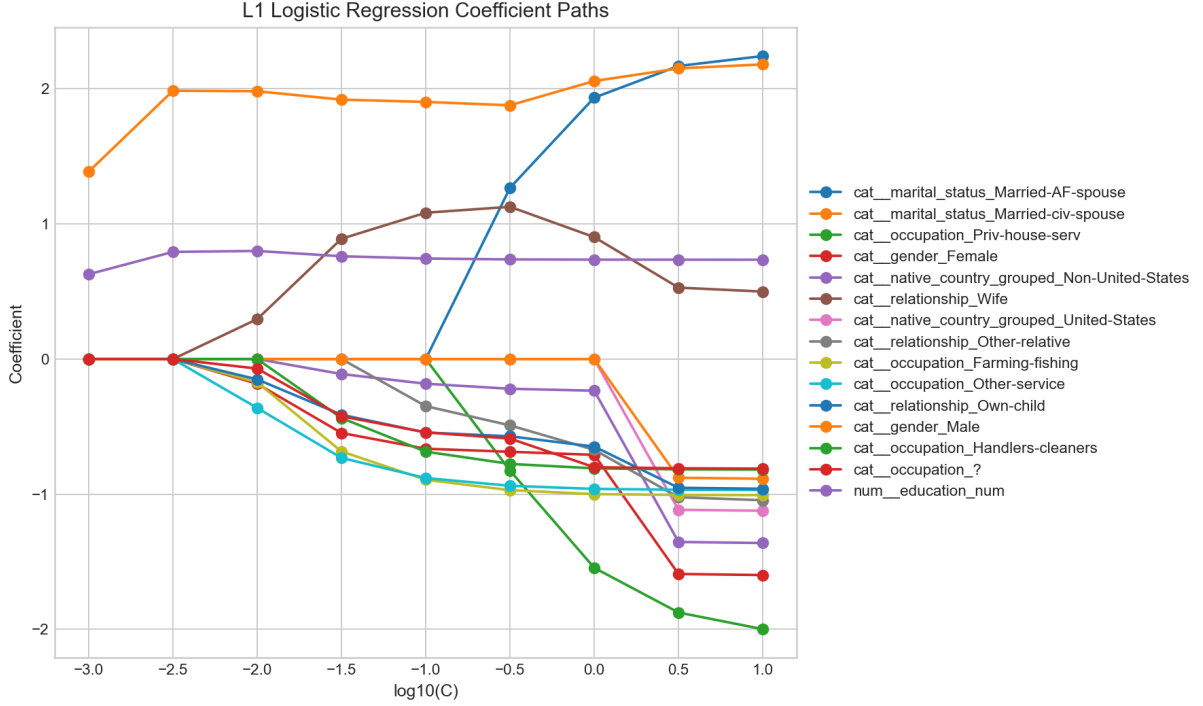


Figure 9: ClawReg coefficient paths across the L1 regularisation grid (ClawLoss). A compact set of feature groups remains active across the full range of regularisation strengths. The scale of the `marital_status` coefficient reflects both predictive signal and the one-hot encoding of individual categories.

Model	Best hyperparameters	CV AUC	Test AUC	Test F1
TabbySVM (linear)	$C = 0.1$	0.893	0.897	0.644
PurrPoly (poly-2)	$d = 2, C = 1.0, r = 1.0$	0.898	0.905	0.670
FurRBF (RBF)	$C = 3.0, \gamma = 0.05$	0.892	0.896	0.670

Table 6: PawSVM comparison across TabbySVM, PurrPoly, and FurRBF. PurrPoly has the highest held-out ROC-AUC. FurRBF does not improve over TabbySVM in this setting.

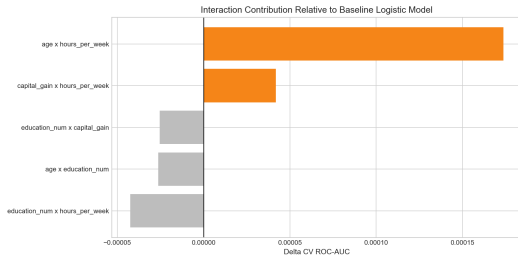


Figure 10: WhiskerInteract: incremental change in CV ROC-AUC from adding each interaction term. All effects are below the 0.001 threshold.

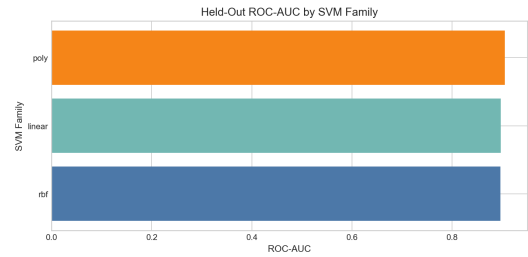


Figure 11: PawSVM held-out ROC-AUC comparison. PurrPoly (degree-2 polynomial) has the highest ROC-AUC among the tested kernels, suggesting modest low-order nonlinearity.

5.7 PurrRobust Stability Checks

Table 7 reports held-out test ROC-AUC for the three main models across random seeds 7, 42, and 99. The ranges are narrow: ClawReg spans 0.897–0.901, TabbySVM spans 0.897–0.901, and PurrPoly spans 0.905–0.906. PurrPoly remains above TabbySVM at every seed, with margins of

+0.0063, +0.0078, and +0.0045 at seeds 7, 42, and 99, respectively.

The TailMI discretisation sensitivity check points in the same direction. The top-10 MI pairs have complete overlap (10/10) across the two binning schemes, and the candidate interaction short-list also has complete overlap (5/5). Figure 12

Model	Seed 7	Seed 42	Seed 99	Range
ClawReg	0.901	0.897	0.901	0.004
TabbySVM	0.900	0.897	0.901	0.004
PurrPoly	0.906	0.905	0.906	0.001

Table 7: PurrRobust seed stability. Held-out test ROC-AUC across three stratified random splits. PurrPoly leads at every seed, and all models show narrow variation across splits.

shows the metric ranges across seeds.

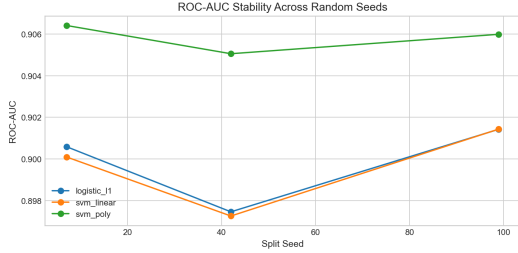


Figure 12: PurrRobust: ROC-AUC metric ranges across seeds 7, 42, and 99. The ordering of PurrPoly above TabbySVM and ClawReg is stable across the three splits.

6 Evaluation Metrics Used

Taken together, the six CatNet modules give a consistent picture of the Adult income dataset. The clearest signals come from household-structure and education-related variables. This pattern appears in the TailMI feature-label ranking, the KittenGauss MeowRank scores, and the ClawReg stability analysis. We interpret this cross-method agreement as useful evidence because the methods make different assumptions and fail in different ways.

The comparison between TailMI and ClawReg adds an important qualification. relationship has the highest feature-label MI (0.116), but it is less stable in ClawReg than marital_status, whose feature-label MI is 0.111. One possible explanation is that the two variables encode overlapping information about household role. Once marital_status is included in the model, relationship contributes less additional signal. This interpretation is consistent with their large pairwise MI of 0.725. The broader implication is that MI rankings should not be read as multivariate feature-importance rankings. High marginal relevance does not necessarily imply low redundancy.

The capital variables illustrate why it helps to compare several views of the same data. Under KittenGauss, capital_gain and capital_loss look poorly described by Gaussian summaries, and

their single-feature MeowRank scores are low. After the WhiskerFlow log1p transform, however, capital_gain_log1p and capital_loss_log1p both survive the full ClawReg regularisation path. This suggests that the apparent weakness of these variables is partly a representation issue. A feature can look uninformative under one summary while still mattering after a more suitable transformation.

The PawSVM boundary analysis addresses the fourth research question. The results suggest that the income decision boundary is not purely linear, but that the nonlinearity is modest. Degree-2 PurrPoly improves ROC-AUC over TabbySVM by roughly 0.008 across the tested seeds, while FurRBF does not improve over the linear baseline. This points toward low-order polynomial structure rather than highly local boundary complexity. The result is also consistent with the WhiskerInteract finding that the five explicit numeric products provide negligible lift. PurrPoly may be capturing a broader set of small second-order effects across the full one-hot encoded feature space.

7 Final Conclusions

This project used CatNet to study the UCI Adult income task through six complementary analyses: WhiskerFlow preprocessing, KittenGauss class-conditional distribution analysis, TailMI feature-dependency diagnostics, ClawReg sparse logistic regression, WhiskerInteract interaction validation, and PawSVM kernel comparison. The goal was not only to obtain a strong classifier, but to understand which parts of the data structure explain the observed performance.

The main findings are consistent across the modules. A compact feature core, including marital_status, education_num, capital_gain_log1p, hours_per_week, age, and capital_loss_log1p, remains stable in the sparse logistic model. Strong TailMI dependencies among household-structure variables appear to reflect redundancy rather than useful numeric interactions. The log-transformed capital variables retain predictive signal even though the raw variables are poorly described by Gaussian summaries. Finally, the PawSVM comparison suggests modest low-order nonlinearity: degree-2 PurrPoly improves over the linear baseline, while FurRBF does not. PurrRobust checks show the same qualitative picture across three random seeds, nine regularisation levels, and two MI

discretisation schemes.

Limitations. KittenGauss provides useful descriptive comparisons, but it is a poor model for zero-inflated variables such as `capital_gain`. Its results should therefore be interpreted alongside ClawReg rather than in isolation. WhiskerInteract tests only five numeric interaction products and does not explore categorical or mixed-type interactions, which could in principle behave differently. PawSVM shows evidence of modest nonlinearity, but it does not identify which specific polynomial terms account for the PurrrPoly improvement. The original pipeline also preserves unknown-value markers as explicit categorical levels rather than imputing or removing affected rows, which may affect categorical frequency summaries. Finally, all conclusions are specific to the UCI Adult dataset and the 1994 United States census population it represents. They should not be generalised to other income prediction settings without independent validation.

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